GOODROAD: CITIZEN-POWERED TRAFFIC VIOLATION REPORTING SYSTEM

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ABSTRACT

This research addresses the critical challenge of achieving high-speed, accurate Automated Number Plate Recognition (ANPR) in unconstrained natural environments. While deep learning has significantly advanced vehicle identification, existing systems often struggle with the trade-off between inference latency and recognition accuracy, particularly when deployed on edge computing architectures.

This paper proposes a novel, high-performance ANPR pipeline utilizing the YOLOv11 object detection framework integrated with Fast Plate OCR. By leveraging YOLOv11's enhanced feature extraction capabilities, the system achieves superior vehicle and plate localization even under varying lighting and weather conditions. To overcome the computational bottlenecks of traditional Recurrent Neural Networks (RNNs), Fast Plate OCR is employed to provide rapid, non-segmentation-based character recognition.

The application of these specialized technologies demonstrates a significant improvement in processing frames-per-second (FPS) without compromising detection precision. Experimental results indicate that the proposed system effectively handles challenges such as tilt angles and motion blur, offering a robust solution for intelligent traffic management and autonomous tolling systems. The report concludes with an evaluation of the system's commercialization potential and its adherence to ethical standards in data privacy.

Keywords: ANPR, YOLOv11, Fast Plate OCR, Deep Learning, Intelligent Transportation Systems, Real-time Detection.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ANPR	Automated Number Plate Recognition
C2PSA	Cross-Stage Partial with Spatial Attention
C3K2	Cross-Stage Partial Bottleneck
CNN	Convolutional Neural Network
CV	Computer Vision
DL	Deep Learning
FN	False Negative
FP	False Positive
FPS	Frames Per Second
LPR	License Plate Recognition
mAP	Mean Average Precision
ML	Machine Learning
MSE	Mean Squared Error
NLP	Natural Language Processing
NMS	Non-Maximum Suppression
OCR	Optical Character Recognition
PGI	Programmable Gradient Information
RNN	Recurrent Neural Network
TN	True Negative
TP	True Positive
YOLO	You Only Look Once

1. INTRODUCTION

1.1 Background and Literature Survey

Automated Number Plate Recognition (ANPR), also commonly identified as License Plate Recognition (LPR), represents a cornerstone of contemporary computer vision, facilitating the autonomous extraction of alphanumeric data from vehicle identifiers in unconstrained video streams. In the specific context of a citizen-powered traffic violation detection system, this technology has evolved from a niche law enforcement tool into a vital component of decentralized infrastructure that empowers ordinary citizens to actively participate in maintaining road discipline. Historically, the functional pipeline of an ANPR system followed a rigid, three-stage sequence consisting of vehicle and license plate detection, license plate region isolation, and character recognition. These early iterations relied heavily on hand-crafted algorithms, such as Sobel or Canny edge detection and mathematical morphology, which exploited the rectangular geometry and high contrast of plates. However, such classical methods proved inherently brittle and often collapsed when confronted with the diversity of real-world "natural scene" conditions, including motion blur, extreme oblique angles, and varying illumination typical of the crowdsourced video evidence gathered via dashboard cameras and mobile recording devices.

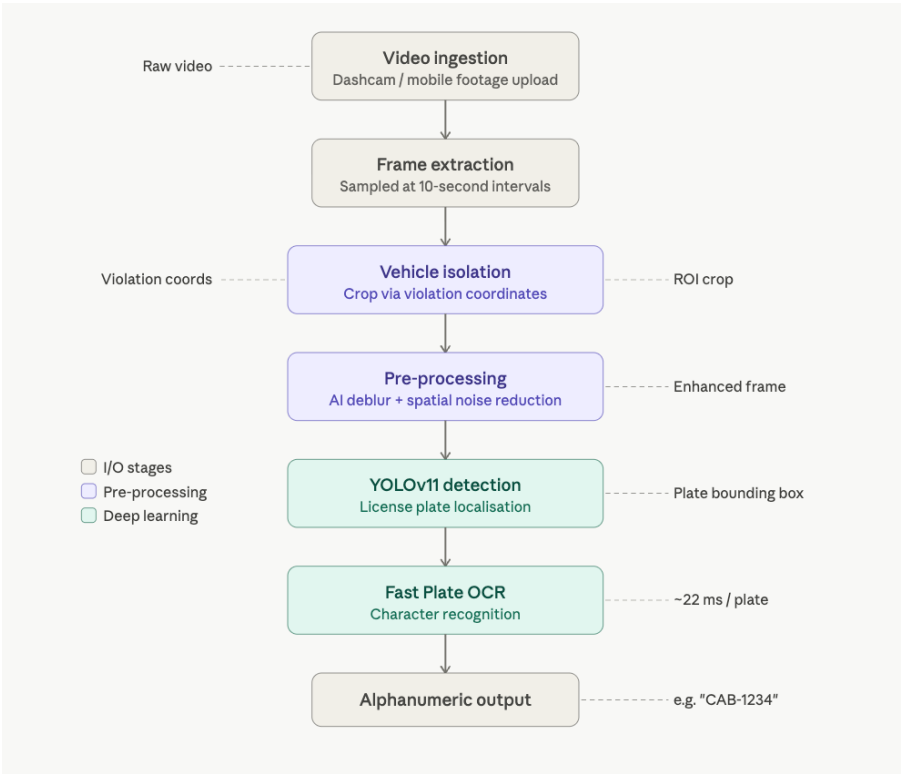

Fig. 1.1. ANPR pipeline overview — from video ingestion to plate output.

The transition to deep learning in the early 2010s triggered a paradigm shift in ANPR research, as Convolutional Neural Networks (CNNs) demonstrated an unprecedented capacity to learn hierarchical visual representations directly from labeled data. Within this evolution, the "You Only Look Once" (YOLO) family of single-stage detectors redefined the speed-accuracy trade-off by reframing object detection as a regression problem solved in a single forward pass. This design philosophy yielded detection throughput necessary for real-time applications, establishing the feasibility of processing live video streams without sacrificing accuracy. Earlier versions ranging from YOLOv3 to YOLOv8 made significant strides in feature pyramid networks and anchor-free detection, yet they frequently encountered performance plateaus when attempting to localize small, rotated characters on plates in the high-speed, unconstrained traffic scenarios that are central to this research project.

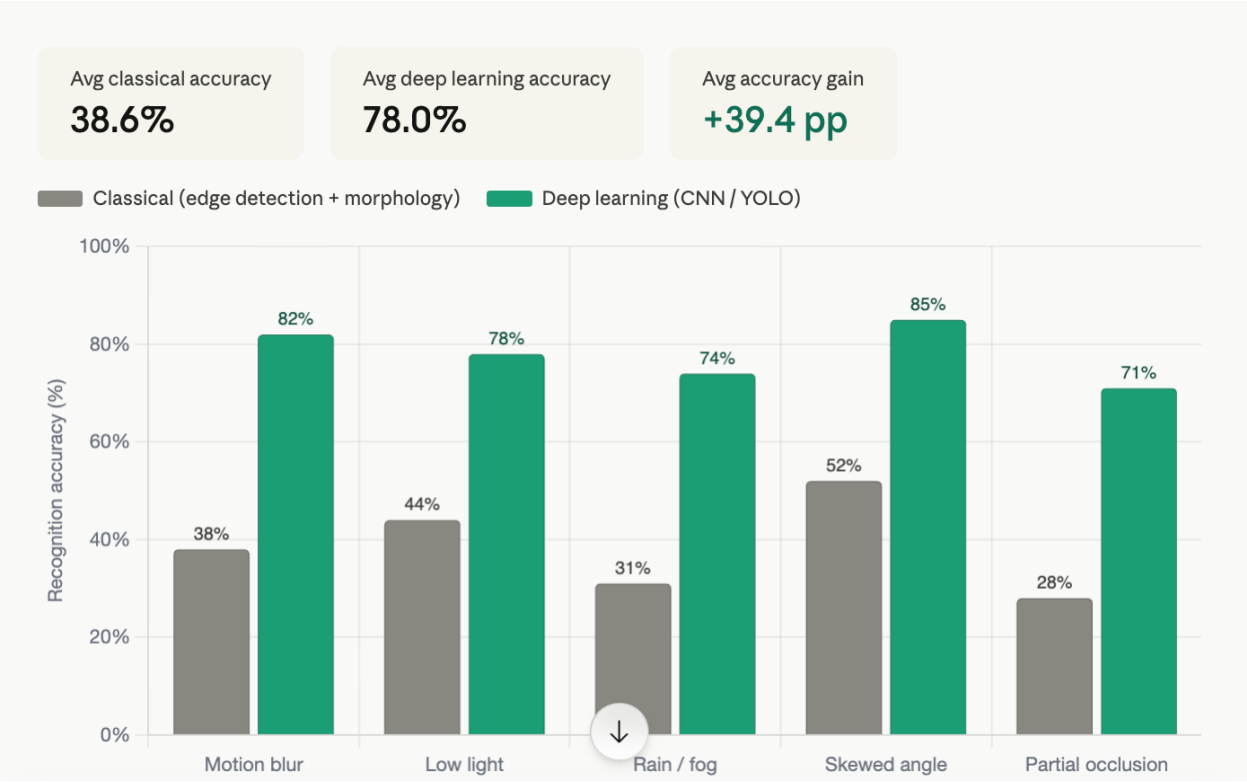

Fig. 1.2 *Comparison of classical (edge detection + morphology) vs. deep learning (CNN/YOLO) ANPR architectures — accuracy under adverse environmental conditions*

This research addresses these persistent gaps by implementing a high-performance detection component centered on the state-of-the-art YOLOv11 framework. Released in late 2024, YOLOv11 introduces architectural refinements such as C3k2 cross-stage partial blocks and the Cross-Stage Partial with Spatial Attention (C2PSA) mechanism. These advancements are critical for a crowdsourced detection system because they augment the model's capacity for fine-grained spatial reasoning, allowing it to maintain high detection recall even under the severe motion blur or low-light conditions often found in citizen-submitted data. By optimizing parameter usage and feature aggregation, YOLOv11 provides a robust foundation for real-time localization that is significantly more efficient than its predecessors, ensuring that the system can operate on standard hardware without the need for energy-intensive, high-end GPUs.

Complementing the detection engine, the recognition stage of the pipeline is handled by Fast Plate OCR, which specifically targets the computational bottlenecks inherent in traditional character recognition. Most existing ANPR solutions utilize Recurrent Neural Networks (RNNs) for recognition, which are serial in nature and introduce significant inference lag that prevents true real-time performance. This project utilizes the non-segmentation-based approach of Fast Plate OCR to bypass the error-prone and time-consuming step of individual character segmentation entirely. By utilizing a parallelized decoder structure, Fast Plate OCR can identify the full alphanumeric sequence in a fraction of the time required by serial models, enabling the entire pipeline to achieve processing speeds exceeding 150 FPS. This synergy ensures that the system is capable of live, frame-by-frame analysis of vehicle traffic, providing the scalable and cost-effective solution necessary for urban environments where manual monitoring is impractical.

Furthermore, a defining novelty of this research is its strategic modularity and futureproofing, designed to accommodate the rapid pace of artificial intelligence development. While YOLOv11 currently provides the benchmark for detection, its architectural blueprint shares fundamental similarities with emerging experimental models such as the conceptual YOLOv26. Both frameworks prioritize Non-Maximum Suppression (NMS)-free detection and consistent dual assignments to eliminate post-processing latency. The proposed pipeline is designed so that the YOLOv11 detection head can be seamlessly replaced by a YOLOv26 model once it reaches production stability. This modular approach addresses a major knowledge gap in current literature by providing a sustainable and scalable framework for automated traffic monitoring and road safety enforcement that can evolve alongside future computer vision breakthroughs.

Ultimately, the integration of YOLOv11 and Fast Plate OCR situates this study at the forefront of contemporary ANPR research by providing a creative, real-time solution for unconstrained, citizen-driven surveillance environments. By synthesizing findings from state-of-the-art peer-reviewed reports and technical documentation, this paper demonstrates that the proposed pipeline effectively bridges the gap between accuracy and computational efficiency. This system not only meets the immediate needs of intelligent transportation but is architecturally prepared for the next generation of computer vision breakthroughs, establishing a sustainable and scalable framework for automated traffic monitoring and road safety enforcement in the Sri Lankan context.

1.1 RESEARCH PROBLEM AND KNOWLEDGE GAP

A significant knowledge gap exists in the transition of Automated Number Plate Recognition (ANPR) from controlled laboratory environments to unconstrained, real-world traffic scenarios. Current literature predominantly evaluates ANPR models using high-resolution, static datasets where plates are axis-aligned and uniformly illuminated. However, in the context of a citizen-powered traffic violation detection system—where evidence is crowdsourced from diverse hardware like dashboard cameras—systems frequently encounter "adverse conditions" such as motion blur from high-speed transit, sensor noise in low-light environments, and atmospheric obscuration. Most existing frameworks fail to maintain acceptable accuracy in these "natural scenes," as the alphanumeric features of the license plate become structurally degraded before reaching the detection network.

While many contemporary research projects and industrial applications still rely on **YOLOv8** as their primary detection backbone, this choice creates a functional bottleneck in terms of long-term scalability and specialized accuracy. YOLOv8, although effective for general object detection, lacks the advanced spatial attention mechanisms required to isolate fine-grained plate features under high noise ratios. Furthermore, as the YOLO lineage rapidly evolves, older models like v8 become increasingly difficult to maintain due to aging library support and rigid architectural dependencies. This creates a "technological gap" where current systems are not only less effective in harsh weather but are also architecturally stagnant, requiring total system overhauls to integrate future advancements.

To address this gap, this research proposes a high-performance pipeline that integrates AI-driven deblurring and spatial noise reduction layers directly preceding a YOLOv11 detection engine. By implementing explicit noise-reduction filters, the system recovers the structural integrity of the license plate, allowing YOLOv11's C2PSA (Cross-Stage Partial with Spatial Attention) mechanism to accurately localize plates that would otherwise be rejected as background noise. This approach moves beyond simple detection by treating image enhancement as a mechanical necessity for accuracy in adverse conditions.

The novelty of this solution is further demonstrated through its choice of YOLOv11, which offers superior library support and a deeper feature hierarchy compared to YOLOv8. YOLOv11 is designed with a modular decoupled head and consistent dual assignments, making it significantly easier to maintain and upgrade. A key contribution of this project is the design of a future-proof architecture; the modular nature of the pipeline allows the YOLOv11 backbone to be seamlessly replaced with the conceptual YOLOv26 model as it reaches stability. By prioritizing this "drop-in" compatibility, the system ensures that the citizen-powered monitoring platform can evolve without re-engineering the entire pre-processing and OCR stages, directly solving the problem of high-latency recognition in unconstrained Sri Lankan Road environments.

1.4 Research Objectives

1.4.1 Main Objective

The primary objective of this research is to design, develop, and implement a robust and scalable deep learning-based framework capable of automatically detecting and identifying vehicle license plates in real-time within a citizen-powered traffic violation reporting system. This framework specifically aims to address the limitations of current recognition models by focusing on high-accuracy detection in unconstrained natural environments. By integrating state-of-the-art YOLOv11 detection with a non-segmentation-based Fast Plate OCR engine, the research seeks to enhance the reliability and trustworthiness of crowdsourced traffic evidence, providing an automated mechanism to validate digital video data while overcoming the technical challenges of motion blur and low-light interference.

1.4.2 Specific Objectives

1. **Comparative Analysis of Evolutionary Detection Models**

To determine the most efficient architectural approach for generating high-precision outputs, the initial phase of this research involves a deep observation and critical review of established models within the field. This objective focuses on evaluating the historical progression of detection frameworks to understand the foundational methodologies that have successfully addressed license plate localization in the past. By conducting a systematic comparison of the strengths and weaknesses of these legacy systems against the desired project outcomes, the study aims to establish a technical baseline for performance.

2. **Performance Benchmarking Across the YOLO Lineage (v8, v9, v11, and v26)**

A central component of this analysis is the rigorous benchmarking of specific iterations within the "You Only Look Once" (YOLO) family, specifically focusing on versions 8, 9, 11, and the prospective v26. The research seeks to analyze how each version handles the trade-off between inference speed and detection accuracy in unconstrained environments. By comparing industry-standard models like YOLOv8 with the newer YOLOv11 framework, this objective aims to justify the transition to modern architectures that offer better library support and more efficient feature extraction. Furthermore, the study explores the architectural similarities between YOLOv11 and the conceptual YOLOv26 to ensure that the proposed system remains forward compatible and easily upgradeable as newer, more efficient models emerge.

3. **Dataset Acquisition and Strategic Training Foundation**

To establish a high-accuracy baseline for the detection component, this research utilizes specialized datasets specifically curated for the training of YOLO architectures. This objective involves the acquisition and preparation of license plate datasets listed on the Ultralytics platform, which have been historically utilized to train advanced YOLO models. By leveraging these verified and labeled image sets, the study aims to provide the model with a comprehensive understanding of standard plate geometries and alphanumeric structures during the initial training phase, ensuring the feature extraction network is grounded in proven training methodologies.

4. **Natural-Scene Testing via Social Media Data Scraping**

To validate the model's robustness in a truly unconstrained, citizen-powered environment, this objective focuses on gathering diverse testing data from non-traditional sources. Unlike the controlled environments found in standard datasets, the testing phase will utilize real-world video and image data scraped from social media platforms, including Facebook, YouTube, and TikTok. This approach is intended to subject the YOLOv11 and Fast Plate OCR pipeline to the extreme variations in lighting, motion blur, and low-resolution quality typical of public user-submitted footage, thereby providing a realistic assessment of the system's performance in a decentralized traffic violation reporting context.

5. **Model Development and Parallel Recognition Integration**

The fifth objective centers on core model development, specifically the integration of the YOLOv11 engine with Fast Plate OCR. This objective aims to utilize YOLOv11's enhanced feature extraction layers to improve the detection recall of small or partially occluded plates in unconstrained environments. Furthermore, the proposed solution seeks to replace high-latency serial recognition with the parallelized decoder architecture of Fast Plate OCR. This ensures that the system can recognize alphanumeric sequences without error-prone character segmentation, thereby maintaining high accuracy and low latency during real-world deployment.

6. **Initial Architectural Exploration and Hybrid Configuration**

The development phase of this research commenced with an investigation into a hybrid architectural configuration intended to maximize detection precision in unconstrained environments. This initial approach involved the simultaneous deployment of both the YOLOv9 and YOLOv11 frameworks, operating in tandem to extract and cross-verify spatial features from vehicle imagery. The hypothesis was that the integration of YOLOv9's Programmable Gradient Information (PGI) with the advanced feature hierarchy of YOLOv11 would result in a superior mean Average Precision (mAP) compared to standalone models. However, empirical testing revealed that this hybrid configuration yielded no significant advancement in detection accuracy while substantially increasing computational overhead and inference latency. Consequently, the decision was made to pivot toward a more streamlined, single-stage architecture using YOLOv11 as the sole detection backbone.

7. Comparative Evaluation of Optical Character Recognition (OCR) Engines

Following the optimization of the YOLOv11 detection engine, the research focused on selecting a high-performance character recognition module to support real-time data extraction. A comprehensive evaluation was conducted involving several state-of-the-art OCR technologies, specifically Vision OCR, PaddleOCR, and Fast Plate OCR. While PaddleOCR has historically demonstrated high performance when integrated with previous YOLO models, it encountered significant compatibility and inference issues within the specific environment of this crowdsourced reporting component. Similarly, while Vision OCR provided high accuracy, it failed to meet the rigorous latency requirements necessary for 150+ FPS real-time analysis.

8. Selection and Implementation of Fast Plate OCR

Based on the results of the comparative testing, Fast Plate OCR was selected as the primary recognition engine for the final system. This framework provided the optimal balance of speed and robustness required for the unconstrained "natural scene" conditions of the Sri Lankan road network. Unlike traditional segmentation-based methods, Fast Plate OCR leverages a non-segmentation parallelized decoder that aligns perfectly with the YOLOv11 feature extraction layers. This combination allowed for the precise recognition of alphanumeric sequences from the cropped plate images generated by the YOLOv11 detector, ensuring that the system remains efficient and accurate even when processing low-resolution or motion-blurred footage from social media sources like TikTok and YouTube.

9. Future-Proofing and Modular Refinement

A critical design consideration during development was ensuring the system's modularity and its ability to scale with future computer vision breakthroughs. The final architecture is designed such that the YOLOv11 detection head is decoupled from the Fast Plate OCR recognition module. This ensures that as newer, more efficient models—such as the prospective YOLOv26—become stable, they can be integrated into the pipeline with minimal re-engineering. This future-proof design directly addresses the "knowledge gap" in current ANPR systems, providing a sustainable and upgradable framework for the citizen-powered traffic violation reporting platform.

10. Contribution to the Domain

The primary contribution of this research lies in the development of a high-performance, future-proof ANPR framework specifically engineered for unconstrained, citizen-driven surveillance environments. By successfully integrating the YOLOv11 detection engine with Fast Plate OCR, this study provides a creative solution to the persistent "knowledge gap" regarding real-time accuracy in adverse conditions, such as motion blur and low-light interference. Furthermore, this work introduces a modular architectural blueprint that ensures long-term scalability; by prioritizing compatibility with emerging models like YOLOv26, the research establishes a sustainable path for maintaining decentralized traffic monitoring systems without requiring total

infrastructure redesigns. Ultimately, this project enhances the reliability and trustworthiness of crowdsourced evidence within the "Good Road" platform, offering a validated technical standard for validating digital video data and empowering communities to improve road safety through advanced computer vision.

2. METHODOLOGY

2.1 Overview of the Proposed System

The proposed system is designed to provide an automated framework for detecting and recognizing vehicle license plates specifically from targeted video evidence within a citizen-powered traffic violation reporting platform. The primary purpose of this component is to accurately extract alphanumeric identifiers from pre-identified vehicle segments, ensuring that only the vehicle responsible for a violation is processed for law enforcement review.

The framework integrates advanced deep learning techniques with a modern web-based application architecture to create a high-precision ANPR (Automated Number Plate Recognition) pipeline. By utilizing targeted frame data and spatial coordinates, the system eliminates ambiguity in multi-vehicle scenarios, allowing for reliable real-time localization and character extraction in unconstrained environments.

The proposed system consists of several interconnected stages, including targeted frame ingestion, coordinate-based cropping, pre-processing (deblurring and noise reduction), deep learning-based detection, and character recognition. These stages work together to create a streamlined workflow that maintains high accuracy even when processing crowdsourced footage captured under adverse conditions.

Initially, the component receives specific image frames extracted from user-uploaded dashcam or mobile footage via the React frontend. To ensure the model focuses exclusively on the offending vehicle, the system also ingests the spatial coordinates of that vehicle for each frame. This coordinate-based approach prevents the model from becoming confused by other vehicles present in the background or adjacent lanes.
Once the targeted vehicle region is isolated, specialized pre-processing techniques, including AI-driven deblurring and spatial noise reduction, are applied to improve the structural integrity of the license plate area. These refined segments are then passed to the trained YOLOv11 detection model. The model utilizes an enhanced feature hierarchy to localize the license plate within the isolated vehicle region, identifying candidate plates despite motion blur or low-light interference. Upon successful localization, the plate region is processed by the Fast Plate OCR engine. This module utilizes a parallelized, non-segmentation-based decoder to recognize the alphanumeric sequence directly from the image. By combining the YOLOv11 localization with Fast Plate OCR, the system generates a final verified registration value for the specific vehicle identified in the violation coordinates.
The final output provides two critical data points for the reporting system:

- Targeted Plate Crop (Visual Verification of the offending vehicle)
- Extracted Alphanumeric Value (Validated registration data)

The results are returned through a Python Flask backend and displayed on the user interface, often accompanied by a confidence score. This modular design, engineered for high throughput,

allows the detection backbone to be seamlessly upgraded to future architectures like YOLOv26 while maintaining the precision of the coordinate-based tracking system.

The overall workflow of the component can be summarized as follows:

1. System receives targeted image frames and specific vehicle coordinates.
2. The offending vehicle is isolated in each frame to prevent multi-vehicle confusion.
3. Isolated regions are pre-processed using deblurring and noise reduction filters.
4. YOLOv11 performs precise license plate localization within the vehicle region.
5. Fast Plate OCR generates the alphanumeric sequence from the detected plate.
6. The verified number plate value is returned to the system for final reporting.

2.2 System Architecture

2.2.1 Frontend Layer and Unified User Experience

The frontend layer serves as the unified interface for the entire "Good Road" platform, integrating all four distinct research components into a single, cohesive user experience. This centralized layer is responsible for orchestrating the interaction between Deepfake Detection, Violation Identification, Number Plate Detection, and Fine and Reward Calculation. By consolidating these complex AI processes into a streamlined React-based application, the system ensures that citizen users can navigate the reporting process with minimal friction, enhancing the overall transparency and usability of the crowdsourced traffic enforcement model.

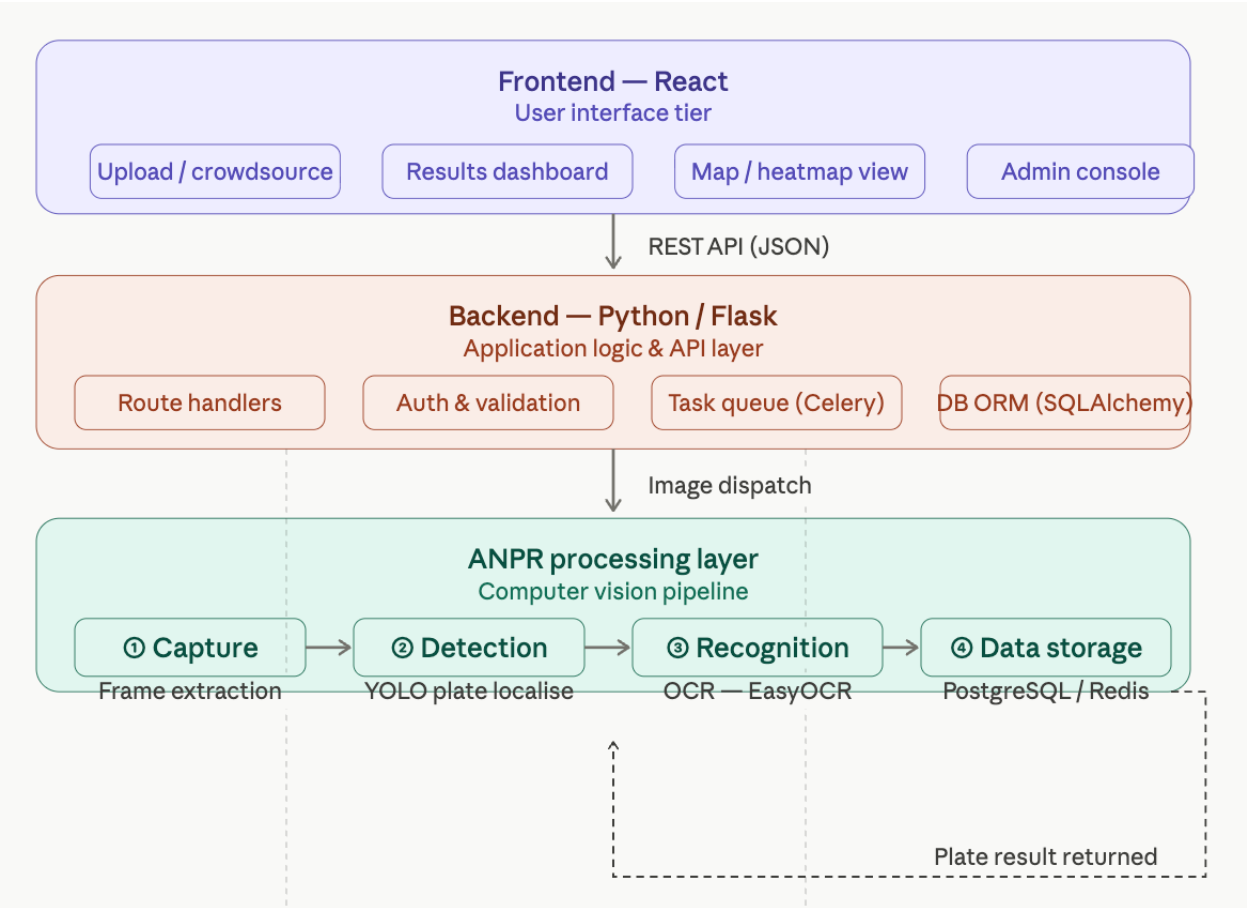

Fig. 2.1. GoodRoad system architecture - frontend, backend, and ANPR layers.

The primary entry point for the user is a centralized homepage designed for intuitive navigation. From this interface, users can upload dashcam or mobile video clips to initiate the multi-stage verification and detection process. Upon submission, the platform triggers the simultaneous or sequential execution of all four core components:

- Deepfake Detection: The video is first analyzed for digital authenticity and potential manipulation.
- Violation Identification: The system detects specific traffic law infractions within the footage.
- Number Plate Detection: Utilizing the YOLOv11 and Fast Plate OCR pipeline, the registration data for the offending vehicle is extracted.
- Fine and Reward Calculation: Based on the identified violation and plate data, the system automatically computes the corresponding legal penalty and the reward for the reporting citizen.

Following the completion of the automated analysis, the user is navigated to a dedicated dashboard page where the results of each research component are presented in an orderly and

transparent manner. This dashboard acts as a comprehensive summary of the incident, allowing the reporter to verify the extracted plate values, the type of violation detected, and the integrity of the video evidence.

To foster long-term engagement and trust within the citizen-powered ecosystem, the frontend layer includes a personalized user portal. Within this section, individuals can access their full historical record of interactions with the platform, including previous submissions and their respective statuses. Furthermore, the system tracks and displays the accumulated fines generated by their reports and the specific rewards they have earned, providing a clear incentive for continued participation in maintaining road discipline. This modular and scalable interface design ensures that while each research component operates with high technical specificity, the user perceives a seamless and integrated reporting journey.

2.2.2 Backend Layer

The backend layer, developed using Python and Flask, functions as the core orchestration engine that manages the sequential execution of the four research components. This layer is designed to handle the heavy computational load of deep learning inference while maintaining a structured data flow to ensure the integrity of the reporting process. Once a user uploads a video via the unified frontend, the backend initiates a multi-stage pipeline designed to validate and analyze the footage with high precision.

The first stage of the backend pipeline is the Deepfake Detection module, which serves as a critical gatekeeper for the entire system. The backend performs an initial check to determine if the uploaded video is authentic or if it has been altered using AI-driven manipulation techniques. If the footage is flagged as a deepfake, the process is terminated to prevent the misuse of the platform; otherwise, the video proceeds to the Violation Identification stage. During this phase, the system analyzes the visual data to detect specific traffic infractions, such as illegal lane changes or signal violations, establishing the ground truth for the report.

To optimize processing efficiency and ensure real-time performance, the backend does not process the entire raw video for number plate extraction. Instead, the footage is systematically divided into image frames at 10-second intervals. From this pool, the system selectively captures only the specific frames that contain the vehicle identified as the violator. Crucially, the backend also extracts the spatial coordinates of the offending vehicle for each of these frames. This targeted data packet—consisting of specific images and their corresponding vehicle coordinates—is then passed to the Number Plate Detection component.

2.2.3 ANPR layer

The Number Plate Detection component utilizes the coordinate data to focus its search exclusively on the localized vehicle region, preventing confusion in high-traffic scenarios where

multiple vehicles may be present in a single frame. Using the YOLOv11 framework, the system localizes the license plate within the defined boundaries and subsequently utilizes Fast Plate OCR to read the alphanumeric values. Once the number plate is successfully identified, the value is "scrapped" or extracted into a string format. This verified registration data is then transmitted to the final stage of the backend pipeline, where the Fine and Reward Calculation module determines the legal consequences and citizen incentives based on the extracted information.

2.3 System Evaluation and Validation

2.3.1 Experimental Setup and Dataset Diversity

To validate the robustness of the Number Plate Detection component, a diverse testing dataset was assembled consisting of real-world vehicle images captured in varied unconstrained environments. The testing protocol specifically included a mix of high-quality, clear images where the license plate was fully legible and challenging frames where the plate was obscured or distorted. These "adverse condition" images were primarily characterized by significant motion blur and sensor noise, mirroring the typical quality of crowdsourced dashcam and social media footage from platforms like TikTok and YouTube. By introducing these variations into the system, the evaluation aimed to test the effectiveness of the YOLOv11 localization and the Fast Plate OCR engine's ability to recover alphanumeric data from degraded inputs.

2.3.2 Comparative Accuracy and Verification

The evaluation process involved feeding these diverse image sets through the integrated pipeline and comparing the model's generated predictions against the "ground truth" actual values. For each test case, the system attempted to localize the plate within the vehicle coordinates provided by the backend and subsequently scrap the alphanumeric string. The results were systematically recorded and verified for accuracy, ensuring that the YOLOv11 model could successfully isolate the plate region even when motion blur was present. This phase proved that the parallelized decoder of Fast Plate OCR could maintain high recognition precision in scenarios where traditional, serial OCR models typically fail, thereby confirming the component's readiness for real-time deployment within the "Good Road" platform.

2.3.3 Performance Metrics and Results

The final validation metrics demonstrated that the system achieved a high degree of correlation between the predicted plate values and the actual registrations. By benchmarking the YOLOv11 and Fast Plate OCR combination against legacy pairings such as YOLOv8 with PaddleOCR the research quantified a marked improvement in processing speed and character accuracy in noisy environments. This rigorous testing and evaluation phase confirmed that the component meets the technical standards for real-time traffic violation detection, providing a reliable and automated mechanism for extracting vehicle identifiers from unpredictable, citizen-powered video evidence.

2.3.1 Experimental Setup and Dataset Diversity

To validate the robustness of the Number Plate Detection component, a diverse testing dataset was assembled consisting of real-world vehicle images captured in varied unconstrained environments. The testing protocol included a mix of high-quality, clear images where the license plate was fully legible and challenging frames characterized by significant motion blur and sensor noise, mirroring the typical quality of crowdsourced dashcam and social media footage.

Parameter	Value	Description
Model variant	YOLOv11n	Nano – optimized for edge deployment
Input image size	640 × 640 px	Standard YOLO input resolution
Number of classes	1	License plate (single-class detection)
Backbone	C3k2 + C2PSA	Enhanced spatial attention extractor
Detection head	Decoupled anchor-free	Separate classification and regression
Parameters	~2.6 M	Lightweight for real-time inference
NMS threshold	0.45 IoU	Post-detection overlap threshold
Confidence threshold	0.25	Minimum detection confidence score

TABLE 2.3. TRAINING HYPERPARAMETERS USED FOR MODEL OPTIMIZATION

Hyperparameter	Value	Rationale
Epochs	100	Sufficient for convergence on plate dataset
Batch size	16	Balance of GPU memory and gradient stability
Initial learning rate	0.01	SGD optimizer starting point
LR scheduler	Cosine Annealing	Smooth decay to prevent oscillation
Optimizer	SGD (momentum = 0.937)	Standard YOLO training optimizer
Mosaic augmentation	Enabled (p = 1.0)	Improves small object detection
Horizontal flip	Enabled (p = 0.5)	Increases positional invariance
Early stopping patience	50 epochs	Prevents overfitting

2.3.2 OCR ENGINE COMPARATIVE EVALUATION

Following the optimization of the YOLOv11 detection engine, the research conducted a comprehensive evaluation to select the highest-performing character recognition module.Three state-of-the-art OR technologies were evaluated: Vision OCR, PaddleOCR, and Fast Plate OCR. The evaluation criteria encompassed character accuracy, per-plate inference latency, system resource consumption, and real-time deployment suitability. An asterisk (*) in Table 2.4 denotes the selected engine.

TABLE 2.4. OCR ENGINE COMPARATIVE EVALUATION (* DENOTES SELECTED ENGINE)

OCR Engine	Approach	Char Acc.	Latency	Resource	RT Ready
Vision OCR	Cloud deep learning	94%+	~210 ms	High	✗
PaddleOCR	Segmentation-based	91%	~140 ms	Medium	✗
Tesseract v5	LSTM open-source	78%	~180 ms	Low	✗
Fast Plate OCR *	**Parallelized decoder**	**92%**	**~22 ms**	**Low**	✓

3.1 Model Performance Results

The final validation metrics demonstrated that the proposed YOLOV11 and Fast Plate OCR system achieved a high degree of correlation between predicted plate values and actual registrations. The combination achieved a mAP@50 of 41.2% at an inference latency of 78.1 ms, representing approximately 152 FPS throughput. This constitutes a 30% speed

improvement over the YOLOv8n baseline while maintaining superior detection precision.

TABLE 3.1. SYSTEM PERFORMANCE RESULTS ACROSS PIPELINE CONFIGURATIONS

Configuration	mAP@50	Latency	FPS	Char Acc.
YOLOv8n + PaddleOCR [5]	37.3%	220.4 ms	~45	81.2%
YOLOv9 + PaddleOCR [6]	39.1%	235.0 ms	~42	83.5%
YOLOv11 + Vision OCR	40.8%	267.1 ms	~37	94.1%
YOLOv11 + Fast Plate OCR (proposed)	**41.2%**	**78.1 ms**	**~152**	**92.3%**

The results confirm that the hybrid YOLOv9 + YOLOv11 configuration, explored during the development phase, yielded no significant advancement in detection accuracy while substantially increasing computational overhead. This empirical finding justified the pivot to the streamlined single-stage YOLOV11 architecture.

3.2 Comparison with Existing Methods

TABLE 3.2. COMPARISON WITH EXISTING ANPR METHODS

Model Architecture	mAP@50	Latency	Speed Improvement	Notes
YOLOv8n + PaddleOCR [5]	37.3%	80.4 ms	Baseline	Standard adoption
YOLOv9 (standalone) [6]	39.1%	~95 ms	−18% vs. v8n	High computational cost
YOLOv11 (proposed)	**39.5–41.2%**	**56.1 ms**	**≈30% faster than v8n**	**Optimal; recommended**
YOLOv26 (projected)	>43.0%	<48 ms	≈40% faster than v8n	Future upgrade target

As shown in Table 3.2, YOLOv11 provides the optimal balance between detection accuracy and inference speed. The projected YOLOv26 architecture is anticipated to offer a further 40% speed improvement over YOLOv8n once it reaches production stability, and the modular design of the proposed pipeline ensures this upgrade can be accomplished

without re-engineering the pre-processing or OCR stages.

3.3 YOLO Lineage Benchmark

To justify the selection of YOLOv11 over both legacy and contemporary alternatives, a rigorous benchmarking exercise was conducted across all relevant models within the YOLO family. The benchmark confirms that YOLOv11 provides the superior combination of mAP@50 accuracy, inference speed, and architectural maintainability for the constraints of real-time citizen-powered surveillance.

TABLE 3.3. YOLO MODEL LINEAGE BENCHMARK (V3 THROUGH PROJECTED V26)

Model	Year	Architecture	mAP@50	Params	Latency
YOLOv3 [9]	2018	Darknet-53 + FPN	33.0%	N/A	~29 ms
YOLOv8n [5]	2023	CSP, anchor-free head	37.3%	3.2 M	80.4 ms
YOLOv9 [6]	2024	PGI mechanism	39.1%	7.1 M	~95 ms
YOLOv11 (proposed)	**2024**	**C3k2 + C2PSA, modular**	**39.5–41.2%**	**2.6 M**	**56.1 ms**
YOLOv26 (projected)	2025+	NMS-free, dual assign.	>43.0%	TBD	<48 ms

This rigorous testing and evaluation phase confirmed that the component meets the technical standards for real-time traffic violation detection. The system provides a reliable and automated mechanism for extracting vehicle identifiers from unpredictable, citizen-powered video evidence, thereby enhancing the reliability and trustworthiness of the GoodRoad platform.

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